You are acting as a senior AI product strategist, public sector consultant, education data scientist, and UX thinker.

We are building a prototype for the **Stay-In School Hackathon** for the School Education Department, Government of Andhra Pradesh.

The goal is not to build another generic student dropout dashboard. The jury will likely see many dashboards showing total students, high-risk students, charts, and filters. We need something more practical, more intelligent, and more aligned with how government officers, district teams, headmasters, and teachers actually work.

Build a **Student Retention Intelligence System** that helps the department reduce school dropouts through early warning, explainability, accountability, and action tracking.

You may choose the best implementation approach for backend architecture, frontend structure, file system, frameworks, or exact UI layout.

1. Product Philosophy

Design the system around one principle:

Every login should answer: “What needs my attention today, and what action should I take?”

The first screen for each user must not be a generic data dump. It must reflect that user’s authority, responsibility, and decision-making power.

A state officer should not first see a long list of individual students.
A district officer should not first see state-level aggregates.
A teacher should not see complex analytics before seeing which students need attention today.

The system should be role-aware, action-first, and drill-down enabled.

2. Available Data Context

Use the uploaded data files:

- Student-level attendance and academic data for 2023-24
- Dropout student ID list for 2023-24
- Dropout student ID list for 2024-25

- School location master data
- Hackathon PDF briefs

Important:

- The detailed full student-level file appears to be available only for 2023-24.
- The 2024-25 file appears to contain dropout IDs only, not the full detailed current-year student file.
- Do not fabricate a full 2024-25 analysis if the data is unavailable.
- Where needed, design future-ready placeholders for 2024-25, LEAP integration, socio-economic data, migration data, and behavioral data.
- Clearly separate what is proven from current data, what is synthetic for demo, and what is future integration-ready.

3. Core System Objective

Build a system that can:

- identify students at risk of dropping out
- explain why they are at risk
- recommend the next best action
- track whether action was taken
- escalate unresolved cases
- show district, mandal, school, class, and student-level risk patterns
- help officers prioritize limited resources
- improve over time through intervention feedback

The system should support both:

1. **Prediction**, who may drop out
2. **Prevention**, what action should be taken now

4. Persona-Specific Capabilities

Design the product around these personas.

A. State-Level Officer View

The state officer is responsible for state-wide outcomes. They are busy and need instant intelligence.

Their home view should help them understand within 5 to 10 seconds:

- which districts need attention
- which districts improved or worsened compared to the previous period
- which districts have the highest concentration of high-risk students

- which districts have poor intervention follow-through
- where escalations are stuck
- which risk drivers are most common across the state
- which districts need academic support, transport support, migration support, or community engagement
- where the department should focus reviews, field visits, or additional resources

Capabilities required:

- district performance comparison
- risk trend by district
- improvement or deterioration since last period
- district ranking by dropout risk burden
- district ranking by intervention completion
- district-level hotspot map
- unresolved escalation tracker
- policy-level insight cards
- ability to drill down from state to district, mandal, school, and student
- ability to issue review notes or action directions to district officers
- ability to see whether directions were acted upon
- scenario view, for example, “If we can intervene in only 10,000 cases this month, where should we focus?”

Do not make the state view only about total high-risk students. Make it about governance, prioritization, and accountability.

B. District Officer View

The district officer is responsible for district-level execution.

Their home view should immediately answer:

- which mandals are performing poorly
- which schools need immediate follow-up
- which headmasters or schools have pending actions
- where absenteeism is increasing
- where academic decline is concentrated
- where parent engagement or home visits are required
- whether previous interventions are working
- which schools need district review this week

Capabilities required:

- mandal-wise risk comparison
- school-wise priority list
- high-risk schools requiring attention

- unresolved cases by school
- intervention completion rate
- teacher/headmaster follow-up status
- mandal or cluster hotspot view
- weekly district action plan
- ability to assign tasks to mandal officers or schools
- ability to review action taken reports
- ability to drill down to school and student details when needed

The district view should be action-heavy, not only analytics-heavy.

C. Mandal / MEO View

The mandal-level officer needs to monitor schools and ensure follow-through.

Their view should answer:

- which schools in my mandal need attention today
- which schools have repeated absenteeism patterns
- which headmasters have not acted on flagged students
- which students need home visits or community follow-up
- what actions are overdue
- what needs to be escalated to district

Capabilities required:

- school-level risk queue
- school follow-up compliance
- attendance deterioration alerts
- home visit queue
- action ageing, pending for 3 days, 7 days, 14 days
- escalation to district officer
- remarks and review logging

D. Headmaster / Principal View

The headmaster is responsible for school-level retention.

Their first screen should answer:

- which students need attention today
- which classes have rising absenteeism
- which teacher needs to follow up
- which parent calls or meetings are pending
- which students have deteriorated recently
- which cases need escalation

Capabilities required:

- school action queue
- class-wise risk view
- teacher-wise pending actions
- student watchlist
- parent outreach tracker
- intervention calendar
- recoverable high-risk list
- students with sudden attendance drop
- students with academic decline
- escalation to mandal officer
- ability to mark action taken
- ability to record intervention outcome

The headmaster should not have to interpret complex analytics before acting.

E. Teacher View

The teacher view should be the simplest and most practical.

A teacher should be able to use it in under 2 minutes.

Their first screen should answer:

- which students should I act on today
- why is each student flagged
- what should I do next
- who needs a parent call
- who needs academic support
- who needs counselling or headmaster escalation
- who has improved after intervention

Capabilities required:

- today's action list
- simple student risk explanation
- next recommended action
- parent call or SMS prompt
- mark action taken
- add teacher observation
- add behavioral or social concern
- flag migration, child labour concern, early marriage concern, financial stress, health issue, lack of parental engagement, or learning difficulty
- update follow-up outcome
- see whether attendance improved after intervention

- Telugu and English language toggle

The teacher should not see stigmatizing labels like “dropout risk” in a way that could harm student dignity. Use positive wording like “needs support,” “attendance support,” or “learning support.”

F. Student / Parent-Facing Possibility

Do not show risk labels to students or parents.

If showing a parent or student view, it should be supportive:

- attendance summary
- marks progress
- school messages
- teacher meeting request
- scholarship or transport support reminder
- positive nudges
- simple “how to support your child” guidance

Parent engagement should be treated as a two-way intervention, not just one-way SMS.

5. LEAP Integration Thinking

Public information suggests LEAP already includes student, teacher, school, attendance, profile, entitlement, MDM, governance, communication, and dashboard-type functions.

We do not currently have API documentation.

So design the prototype as **LEAP-compatible**, not falsely LEAP-integrated.

Show what the system would ideally pull from LEAP:

- student profile
- student ID
- school ID
- class and section
- attendance
- marks
- teacher mapping
- school details
- parent contact if available
- entitlement/scheme data if available
- transport or scholarship indicators if available

Show what the system should ideally push back to LEAP:

- risk score or support priority
- recommended action
- teacher action queue
- parent communication trigger
- intervention status
- teacher observations
- behavioral and social indicators
- escalation status
- intervention outcome

Design a mock integration layer or integration concept that shows:

- what data is needed
- what direction the data flows
- how often it should sync
- which user role can see or update each field
- how data privacy is maintained

Do not invent real API endpoints unless they are clearly marked as illustrative.

6. Intelligence Capabilities

Build capabilities, not just charts.

A. Early Warning Risk Engine

Use available attendance and academic data to identify students who need support.

Risk factors may include:

- low attendance
- recent attendance decline
- long absence streaks
- repeated absence patterns
- low marks
- declining marks
- weak school-level context
- district or mandal hotspot context
- Behavioural indicators that we can capture from Leap app. For illustration purpose we can create a form to teacher fill to understand the basic behavioural parameters if default parameters from leap app are not sufficient.
- demographic or social risk indicators if available or synthetically simulated

Create risk tiers, but ensure they are used carefully:

- Low

- Watch
- High Support Needed
- Critical Support Needed

Avoid punitive language.

B. Synthetic Social and Behavioral Data Layer

Because the current dataset may not include all household, migration, behavioral, or socio-economic data, create a synthetic demo layer to show how the system would work when these signals are available.

Synthetic fields can include:

- seasonal migration possibility
- parent engagement level
- financial stress indicator
- child labour concern
- early marriage vulnerability flag
- behavioral disengagement
- peer isolation
- disability support need
- transport difficulty
- household support level

Clearly mark these as synthetic or assumed demo fields. Ideally These fields can be included in the Leap itself but for demo illustration purpose we can show.

Use them to demonstrate how the system becomes stronger when school observations and welfare datasets are connected.

C. Explainability Engine

Every flagged student should have a plain-language explanation.

Examples:

- “Attendance has fallen sharply in the last 30 days.”
- “Student has repeated absence streaks and declining assessment performance.”
- “Risk is driven more by absenteeism than academic performance.”
- “Student appears recoverable because marks are stable but attendance has recently dropped.”

Different roles should see different explanation depth:

- teacher: simple reason and next action

- headmaster: student and class pattern
- district officer: school and mandal pattern
- state officer: district trend and system-level driver

D. Recoverability Score

Do not only show risk. Add a “recoverability” layer.

This should identify students who are at risk but likely to respond to timely intervention.

For example:

- high risk, high recoverability
- high risk, low recoverability
- medium risk, high recoverability
- watchlist, improving

This is important because departments have limited manpower. They need to know where action can produce the highest impact.

E. Intervention Recommendation Engine

Translate risk into action.

Recommended actions may include:

- teacher check-in
- parent call
- parent SMS
- academic remediation
- counselling
- headmaster meeting
- home visit
- transport support verification
- scholarship/welfare linkage check
- migration verification
- community volunteer follow-up
- escalation to mandal or district officer

Each action should have:

- owner
- timeline
- status
- escalation rule
- outcome tracking

Ideally, a LEAP app exists for students and teachers both and there are communication modules as well. For illustration purposes we can show intervention and track intervention on the dashboard, but we need to make things as simple as possible. Will attach a user manual for Leap app as well for review.

F. Parent Engagement Engine

Parent engagement should not be a checkbox.

Design it as a closed-loop communication workflow:

- identify reason for parent outreach
- generate simple parent-friendly message
- avoid saying “your child is dropout risk”
- send positive, actionable message
- track whether parent responded
- track whether attendance improved afterward
- escalate if no response or no improvement

Examples:

- Attendance issue: ask parent to ensure attendance tomorrow
- Academic issue: invite child to remedial support
- Migration issue: ask family to inform school if travel is planned
- Chronic absenteeism: trigger teacher-parent call or home visit

G. Escalation and Accountability Layer

The system should track actions, not just display risk.

For every flagged student or school, track:

- recommended action
- assigned owner
- due date
- action status
- remarks
- outcome
- escalation status
- days pending

Escalation logic:

- teacher does not act → headmaster notified

- headmaster does not resolve → mandal officer notified
- mandal does not resolve → district officer notified
- district trends worsen → state officer sees hotspot or review alert

This is critical. The system must become an execution tool, not just an analytics tool.

H. Forecasting and Incremental Improvement

Add forecasting capability at multiple levels.

Forecast:

- likely dropout risk next month
- expected high-risk student count by school
- district or mandal risk trend
- intervention load
- schools likely to deteriorate if no action is taken

Design incremental improvement:

- model improves as new attendance comes in
- model improves as teacher observations are added
- model improves as intervention outcomes are captured
- model learns which interventions work for which student profiles
- model recalibrates every month or quarter
- system tracks model drift and fairness

7. Dashboards Should Be Role-Specific, Not Generic

Do not create one dashboard and apply filters for every role.

Create role-specific home views based on responsibility.

State Home View Should Emphasize

- district comparison
- statewide trend
- districts worsening
- districts improving
- districts with poor action completion
- major risk drivers
- policy attention areas
- escalation backlog
- resource allocation priorities

District Home View Should Emphasize

- mandal comparison
- school priority list
- pending school actions
- hotspot clusters
- intervention completion
- this week's action plan
- schools needing review

Mandal Home View Should Emphasize

- school follow-up
- overdue actions
- repeated absenteeism clusters
- home visit queue
- teacher/headmaster action completion

Headmaster Home View Should Emphasize

- today's student action queue
- class-wise risk
- teacher-wise pending follow-up
- parent calls pending
- recoverable high-risk students
- students improving or worsening

Teacher Home View Should Emphasize

- today's 5 to 10 priority students
- why each student needs support
- what action to take
- mark action done
- add observation
- parent communication
- student progress after intervention

8. Drill-Down Logic

Allow every higher-level user to drill down, but do not make drill-down the starting point.

State → District → Mandal → School → Class → Student
District → Mandal → School → Student
Mandal → School → Student
Headmaster → Class → Student
Teacher → Student

Each drill-down should preserve context.

Example:

If state officer clicks a district because it is worsening, the district page should explain why it is worsening, not just show raw lists.

9. Action Tables Required

Create action-oriented tables, not only analytical tables.

State Action Table

- district
- risk trend
- change since previous period
- dominant risk driver
- intervention completion rate
- unresolved escalations
- recommended state action

District Action Table

- mandal
- schools needing attention
- high-risk concentration
- overdue actions
- dominant issue
- district action required

School Action Table

- class
- number of students needing support
- teacher responsible
- pending actions
- students improving
- students worsening

Student Action Table

- student ID
- support priority
- reason
- recommended action
- owner
- due date
- status

- outcome

Recoverability Table

- student
- risk level
- recoverability level
- best intervention
- expected impact
- owner

10. Metrics That Matter by Persona

Do not use the same metrics for everyone.

State Metrics

- district risk index
- change in risk vs previous period
- district intervention completion rate
- high-risk student concentration
- unresolved district escalations
- forecasted dropout burden
- resource requirement estimate

District Metrics

- mandal risk index
- school hotspot count
- pending intervention count
- mandal-level deterioration
- parent engagement completion
- home visit completion
- school accountability score

Mandal Metrics

- school follow-up completion
- overdue cases
- repeated absenteeism cases
- schools needing field visit
- escalation count

Headmaster Metrics

- students needing support today

- class-wise risk
- teacher follow-up completion
- parent calls pending
- attendance recovery after intervention

Teacher Metrics

- students to act on today
- parent calls due
- students absent repeatedly
- students improving
- actions pending

11. Language and Usability

Add Telugu and English switch capability.

The UI should be simple enough for a village teacher and strong enough for a state officer.

Avoid technical terms like “SHAP value” in the normal user interface. Use plain explanations.

Technical explainability can be available in an admin or model audit section.

12. Privacy, Ethics, and Dignity

Design the system responsibly.

Capabilities required:

- anonymized analytics where possible
- role-based access
- student-level access only for authorized users
- no public exposure of sensitive information
- no punitive use of risk scores
- no stigmatizing labels for students or parents
- teacher or counsellor remains final decision-maker
- audit trail for automated recommendations
- bias checks across gender, caste/social category, geography, and school type where data permits

13. Data Analysis Expected

Analyze the uploaded data enough to understand:

- attendance patterns
- academic performance patterns

- dropout-labeled student differences
- school and district mapping
- high-risk concentration
- whether early attendance signals are predictive
- whether school-level patterns exist
- whether dropout risk is clustered geographically

If the full data is large, sample intelligently for exploration but design the system to scale to the full dataset.

Generate insights that can be shown in a pitch demo.

Useful insight types:

- districts or schools with concentrated risk
- attendance deterioration before dropout
- difference between dropout and non-dropout students
- schools where many students show similar risk drivers
- students with high risk but high recoverability
- schools where action backlog is high
- districts where risk is increasing despite interventions

14. Model Requirements

Use appropriate ML methods, but do not make the demo only about model complexity.

Required capabilities:

- baseline model
- stronger model for tabular prediction
- interpretable risk drivers
- risk tiering
- early-warning version using partial-year data
- synthetic social/behavioral enhancement layer
- evaluation using imbalanced classification metrics
- top-risk capture analysis
- fairness and bias check where possible
- periodic retraining logic

Metrics to show:

- recall for actual dropouts
- precision
- F1
- ROC-AUC or PR-AUC
- top 10% risk capture rate

- false positive burden
- early detection performance using first 30 or 60 days if possible

15. Intervention Effectiveness

Even if actual intervention outcome data is not available, design a prototype workflow that can capture it.

Track:

- action recommended
- action taken
- date of action
- owner
- student attendance after action
- academic improvement after action
- parent response
- case closed or escalated

Show how this creates a feedback loop.

Over time, the system should learn:

- which intervention works for which risk profile
- which schools are acting consistently
- which districts need administrative support
- which students recover after early intervention

16. Integration With Other Data Sources

Design the system to become stronger when connected with additional data.

Future data sources:

- LEAP attendance
- LEAP marks
- LEAP student profile
- school master
- GIS/school location
- welfare/scholarship data
- transport allowance data
- ration card usage or migration proxy
- teacher observations
- parent communication records
- community worker follow-up

- disability support data
- household socio-economic indicators

For each source, show:

- what it adds
- how it improves prediction
- how it improves intervention
- what privacy care is needed

17. What Not To Do

Avoid:

- generic dashboard for all roles
- too many charts without actions
- student lists at state level as the first view
- black-box risk scores
- overclaiming with synthetic data
- showing sensitive student labels to parents or students
- making teachers do too much extra data entry
- designing for ML accuracy only
- ignoring action tracking
- ignoring escalation
- ignoring LEAP compatibility
- ignoring Telugu usability

18. Final Output Expected

Create a prototype and explanation that demonstrates:

1. What the system does
2. How each role uses it
3. How risk is identified
4. How risk is explained
5. How action is recommended
6. How intervention is tracked
7. How escalation works
8. How LEAP integration would work
9. How the system improves over time
10. How the department can use it to reduce dropouts

Also create a short demo narrative:

- State officer logs in and sees which districts need attention

- District officer sees which mandals and schools require action
- Headmaster sees which students need support today
- Teacher sees exact next steps
- Parent communication is triggered
- Intervention is marked
- System checks whether attendance improved
- Unresolved cases escalate
- The model learns over time

19. Strong Positioning Line

Build the prototype around this idea:

From early warning to verified action, the system helps every level of the department know where dropout risk is rising, why it is rising, who must act, and whether action worked.

20. Final Judging Lens

Optimize the product for a jury that is tired of ordinary dashboards.

The system should feel:

- government-ready
- role-aware
- action-oriented
- explainable
- scalable
- LEAP-compatible
- teacher-friendly
- useful within 10 seconds of login
- strong enough for state leadership
- simple enough for school-level execution

Do not ask the user to make detailed technical decisions unless absolutely necessary. Use your judgment to create the best possible prototype.